

Multivariate Gaussians

A mean point and a covariance ellipse

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Mathematical Foundations for AI · IIT Gandhinagar

Why multivariate Gaussians appear in ML

One word stalks every ML paper you will ever read

You will read...	Where
<i>“weights initialized from $\mathcal{N}(0, 0.02^2)$”</i>	GPT-2's actual init
<i>“add noise $\varepsilon \sim \mathcal{N}(0, I)$ at every step”</i>	diffusion image generators
<i>“sample a latent $z \sim \mathcal{N}(0, I)$”</i>	variational autoencoders
<i>“we assume Gaussian observation noise”</i>	regression, Kalman filters, GPs

Four different corners of AI. One distribution, over and over — never uniform, never exponential.

Why does everyone reach for the same bump?

Because it earned three superpowers

1. **It shows up uninvited.** Add up many small independent effects — anything — and a Gaussian emerges. (The **Central Limit Theorem**; demoed today.)
2. **It assumes the least.** Know only a mean and a covariance? The Gaussian is the honest choice — every alternative smuggles in extra assumptions. (**Max entropy**; stated today, proved in Module 5.)
3. **It never leaves the family.** Slice it, project it, shift it, scale it, add two — the answer is a Gaussian again, with formulas you can write down. (**Closure**; you'll watch it happen today.)

The aristocrat of distributions: born everywhere, presumes nothing, and every operation ML performs keeps it in the family. Today we meet the full multivariate version.

L12 gave us bumps. But data is a vector.

Last lecture: densities for **one** continuous quantity — $p(x)$, area = probability, $\mathcal{N}(\mu, \sigma^2)$.

But since L3, our data has been **vectors**: (height, weight) · a 784-pixel image · an embedding · a parameter vector θ .

- Height and weight are not two separate stories: tall people tend to be heavier. A model of each alone **misses the relationship**.
- We need one density over $\mathbb{R}^d$ — a **joint** — that knows how coordinates co-vary.

Today's object: $x \sim \mathcal{N}(\mu, \Sigma)$ — a bump over $\mathbb{R}^d$.
One point μ says *where*; one matrix Σ says *what shape*.

Learning outcomes

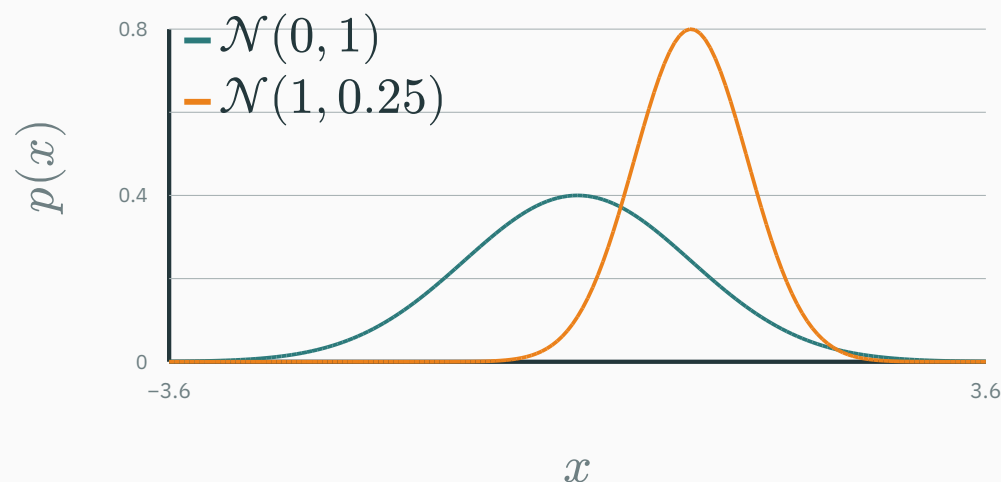
By the end of this lecture you will be able to:

1. Write and **dissect the MVN density** — and say what every symbol is doing there.
2. Read a **covariance matrix** like a picture: variances, correlation, tilt.
3. Compute a 2×2 covariance **by hand** from raw data.
4. **Marginalize** (project) and **condition** (slice) a Gaussian — with formulas.
5. Score outliers with **Mahalanobis distance** — “how many σ ’s, direction-aware”.
6. Prove the ★ fact: **contours are ellipses along the eigenvectors of Σ** (L5, cashed in).
7. Explain why Gaussians are everywhere: **CLT, max entropy, closure**.

From one bump to a 2-D bump

The 1-D bell, recalled (L12)

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad \text{two numbers: center } \mu, \text{ spread } \sigma$$



The whole personality of $\mathcal{N}(\mu, \sigma^2)$ is (μ, σ) . Question: what replaces these two numbers when x becomes a **vector**?

First 2-D density: two independent bells, multiplied

Let x and y be **independent** Gaussians (L12: independence means the joint density factorizes):

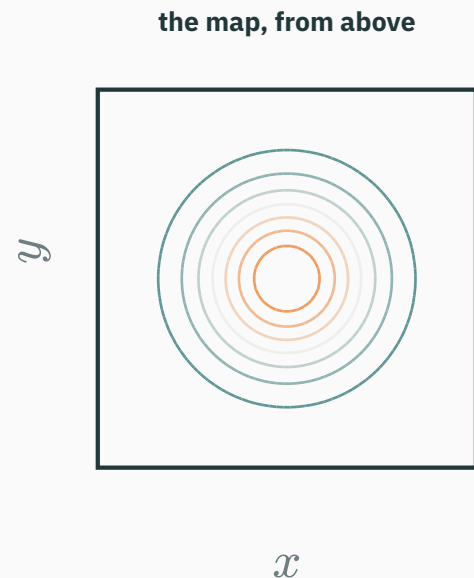
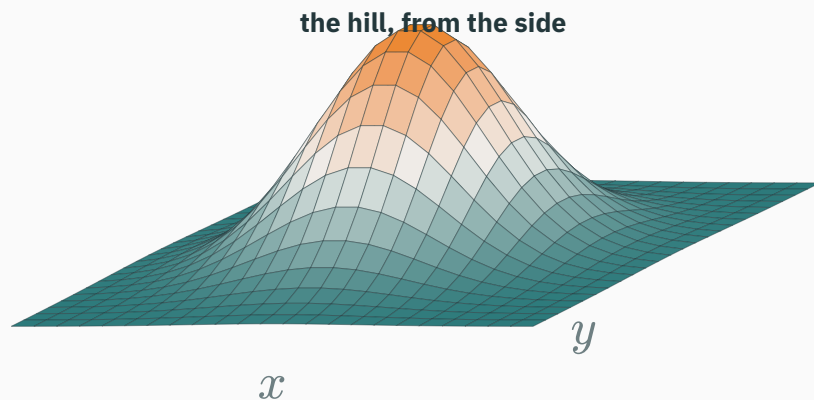
$$p(x, y) = p(x) \cdot p(y) = \frac{1}{\sigma_1 \sqrt{2\pi}} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}} \times \frac{1}{\sigma_2 \sqrt{2\pi}} e^{-\frac{(y-\mu_2)^2}{2\sigma_2^2}}$$

Collect the pieces — constants together, exponents added:

$$p(x, y) = \frac{1}{2\pi \sigma_1 \sigma_2} \exp\left(-\frac{1}{2} \left[\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(y-\mu_2)^2}{\sigma_2^2} \right]\right)$$

A perfectly legal density over $\mathbb{R}^2$ — our first 2-D bump. Every multivariate Gaussian is this, plus one twist we'll add shortly.

The 2-D bump, two views



same object ($\mu_1 = \mu_2 = 0, \sigma_1 = \sigma_2 = 1$), sampled live from its formula — the right view's rings are **contours**: curves of equal density

Cartographers had this figured out long ago: a hill **is** its contour map. We will live in the map view.

Probability is volume now

L12's rule was “probability = **area** under the curve”. One dimension up:

$$P((x, y) \in A) = \int \int_A p(x, y) dx dy$$

probability = **volume** under the hill, above the region A

- Total volume under the whole hill: exactly 1.
- $p(x, y)$ is still a **density**, not a probability: P of any single point is 0, and p can exceed 1 (both L12 facts, still in force).

Contours are the map's way of drawing the hill. Everything today — shape, slicing, projecting — will be read off contour pictures.

Repackage the exponent: a quadratic form appears

Write $x = \begin{pmatrix} x \\ y \end{pmatrix}$, $\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$, $d = x - \mu$. The exponent was a sum of squares — watch it become a **quadratic form** (L9!):

$$\frac{(x - \mu_1)^2}{\sigma_1^2} + \frac{(y - \mu_2)^2}{\sigma_2^2} = d^\top \begin{pmatrix} 1/\sigma_1^2 & 0 \\ 0 & 1/\sigma_2^2 \end{pmatrix} d = d^\top \Sigma^{-1} d$$

where $\Sigma = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{pmatrix}$ collects the two variances — our first covariance matrix, and Σ^{-1} simply divides each squared deviation by its own variance.

L9 planted exactly this: “*the Gaussian’s exponent IS a quadratic form, with $A = \Sigma^{-1}$.*” The IOU starts being paid right here.

Repackage the constant: a determinant appears

The constant out front is secretly a determinant — for a diagonal matrix, $\det =$ product of the diagonal (L4):

$$\frac{1}{2\pi \sigma_1 \sigma_2} = \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2}} = \frac{1}{2\pi \det(\Sigma)^{1/2}}$$

$$p(\mathbf{x}) = \frac{1}{2\pi \det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2} \mathbf{d}^\top \Sigma^{-1} \mathbf{d}\right)$$

nothing was added — the independent case re-packaged itself.

The general formula: keep the packaging, free the matrix

The repackaged formula makes sense for **any** symmetric positive-definite Σ — not just diagonal ones. That generalization **is** the multivariate Gaussian, in any dimension d :

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right), \quad \mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

- $\boldsymbol{\mu} \in \mathbb{R}^d$: the **mean vector** — where the peak sits.
- $\Sigma \in \mathbb{R}^{d \times d}$: the **covariance matrix** — symmetric, positive definite. Off-diagonal entries are the “twist”: they **tilt** the bump, encoding co-variation that independence forbade.

An honest reaction to this formula: *how do we get such a weird-looking thing?! We just did* — it’s the independent case with the matrix let loose. The rest of today makes every piece **visual**.

The formula, dissected

$$p(\mathbf{x}) = \underbrace{\frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}}}_{\text{volume bookkeeping}} \exp \left(-\frac{1}{2} \underbrace{(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})}_{\text{a quadratic form in } \mathbf{x} - \boldsymbol{\mu}} \right)$$

- The **exponent** is L9's quadratic form $\mathbf{d}^\top A \mathbf{d}$ with $A = \Sigma^{-1}$ — L9 planted exactly this: “*the bowl's ellipse contours are a Gaussian's, with $A = \Sigma^{-1}$* ”. Today that IOU is paid.
- The **normalizer** only rescales height so total volume is 1 — $\det(\Sigma)^{1/2}$ is the bump's footprint volume (L4: determinants measure volume). It never affects the shape.
- **Sanity check** $d = 1, \Sigma = (\sigma^2)$: $\det(\Sigma)^{1/2} = \sigma, \Sigma^{-1} = 1/\sigma^2$ — L12's bell, exactly.

Reading the covariance matrix

Σ

The covariance matrix, defined

$$\Sigma_{ij} = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)]$$

- **Diagonal** ($i = j$): $\Sigma_{ii} = \mathbb{E}[(X_i - \mu_i)^2] = \text{Var}(X_i)$ — each coordinate's own spread.
- **Off-diagonal** ($i \neq j$): do X_i and X_j move **together**? Positive: above-average together. Negative: one up, the other down. Zero: no linear relationship.
- Swap i and j : same product, same expectation. So $\Sigma_{ij} = \Sigma_{ji}$ — **every covariance matrix is symmetric**, automatically. (You know what that buys us: L5's spectral theorem. Hold that thought.)

Numbers: four data points, by hand

Four measurements of (x, y) : $(1, 5)$, $(3, 3)$, $(3, 5)$, $(5, 7)$. Mean: $\mu = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$. Deviations $d_i = x_i - \mu$, then average the products:

point	deviation (d_x, d_y)	d_x^2	d_y^2	$d_x d_y$
$(1, 5)$	$(-2, 0)$	4	0	0
$(3, 3)$	$(0, -2)$	0	4	0
$(3, 5)$	$(0, 0)$	0	0	0
$(5, 7)$	$(2, 2)$	4	4	4
mean		2	2	1

$$\Sigma = \begin{pmatrix} \text{mean } d_x^2 & \text{mean } d_x d_y \\ \text{mean } d_x d_y & \text{mean } d_y^2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

the slide recomputes this Σ from the raw points at compile time (chalkdust) ✓

Wait — you have met this matrix before

$$\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

L5's running matrix. You already know its secrets by heart: eigenvectors $(1, 1)$ and $(1, -1)$, eigenvalues 3 and 1. File that away — in the ★ section it becomes a picture.

For any direction a , the variance of the projection $a^\top x$ is

$\text{Var}(a^\top x) = a^\top \Sigma a$ — a quadratic form again, and variances can't be negative...

$a^\top \Sigma a \geq 0$ for every a : covariance matrices are exactly L9's positive semi-definite club. Σ is a machine that turns directions into variances.

Correlation: the unit-free reading

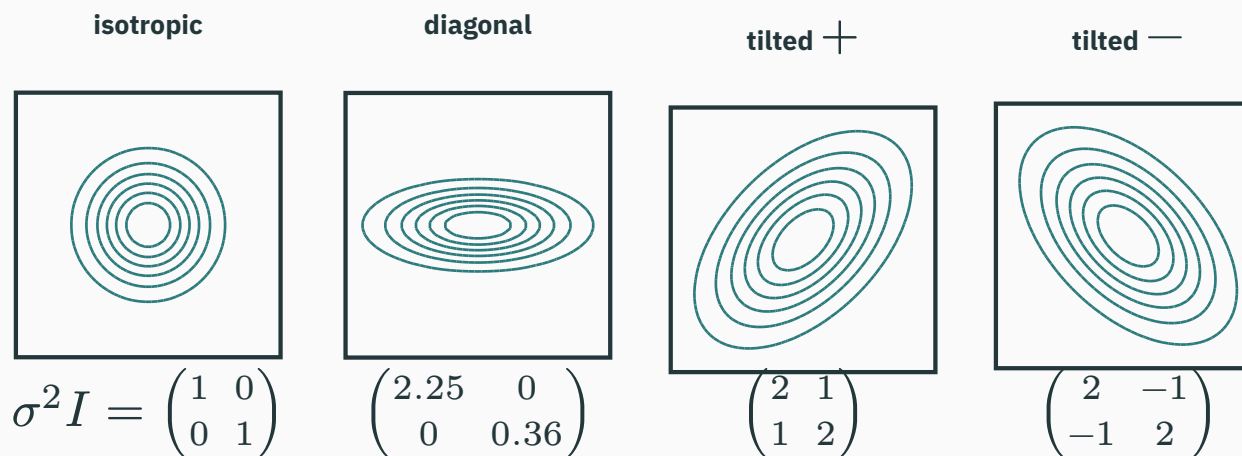
Covariance carries units (cm·kg — unreadable). Normalize by both spreads:

$$\rho = \frac{\Sigma_{12}}{\sigma_1 \sigma_2} = \frac{\Sigma_{12}}{\sqrt{\Sigma_{11} \Sigma_{22}}}, \quad \rho \in [-1, 1]$$

- $\rho = \pm 1$: a perfect line · $\rho = 0$: no **linear** co-movement · sign = tilt direction.
- Our running Σ : $\rho = \frac{1}{\sqrt{2 \cdot 2}} = 0.5$ — moderately, positively coupled.

For **Gaussians** (and only for them, in general): $\rho = 0 \Leftrightarrow$ independent. For arbitrary distributions, $\rho = 0$ does NOT imply independence — a caution flag worth remembering.

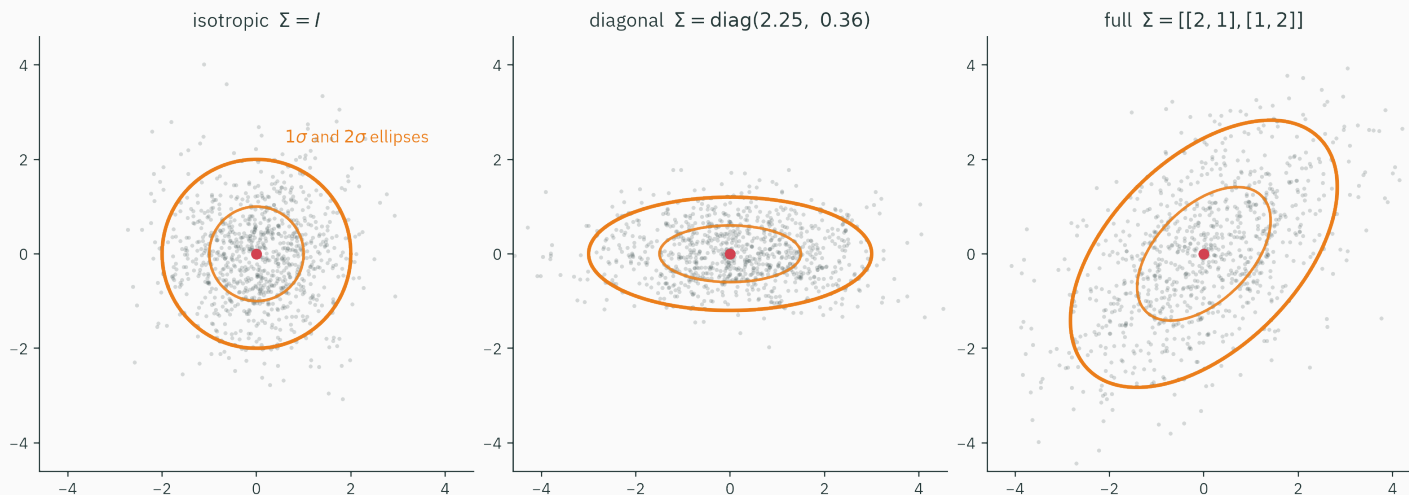
The gallery: what Σ does to the bump



four live densities, one formula — only Σ changed: circles $\rightarrow$ axis-aligned stretch $\rightarrow$ tilt (sign of Σ_{12} picks the tilt's direction)

μ moves the bump. The diagonal of Σ stretches it. The off-diagonal **tilts it.**

Sampled clouds obey the same Σ



900 samples from each density; the 1σ and 2σ ellipses are drawn from Σ **alone**. The cloud fills its ellipse — simulation and formula agree.

The picture to keep: **a Gaussian is a data cloud's idealization** — center μ , shape Σ . Where the ellipse comes from: the ★ derivation, soon.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/multivariate-normal

Drag the entries of Σ (variances and correlation) and watch three things respond at once: the sampled **cloud**, the **contour ellipse**, and the **marginals** on the walls. Then slide the conditioning bar and watch a slice move — the two operations of the next section, live under your mouse.

Five minutes with the sliders builds the reflex: you should **see** $\begin{pmatrix} 4 & -3 \\ -3 & 9 \end{pmatrix}$ as a shape before you compute anything about it.

Code: Σ in one line

```
X = np.array([[1., 5.], [3., 3.], [3., 5.], [5., 7.]])
X.mean(axis=0)           # array([3., 5.])           =  $\mu$ 
np.cov(X.T, bias=True)    # [[2., 1.], [1., 2.]]      =  $\Sigma$  (divide by  $n$ )
np.cov(X.T)               # [[2.67, 1.33], ...]       (divide by  $n-1$ )
```

```
d = torch.distributions.MultivariateNormal(
    loc=torch.tensor([3., 5.]),
    covariance_matrix=torch.tensor([[2., 1.], [1., 2.]])
xs = d.sample((100_000,)) # 100k points from our Gaussian
torch.cov(xs.T)           #  $\approx$  [[2.00, 1.00], [1.00, 2.00]]  $\checkmark$ 
```

`bias=True` divides by n (what we did by hand); the default divides by $n - 1$. Why two conventions exist is an **estimation** question — L14's business, teased next lecture.

$\Sigma = \begin{pmatrix} 4 & -3 \\ -3 & 9 \end{pmatrix}$ describes a data cloud. Which reading is correct?

A. $\sigma_x = 4$, $\sigma_y = 9$, cloud tilted up-right

B. $\sigma_x = 2$, $\sigma_y = 3$, cloud tilted down-right

C. Invalid — a covariance matrix can never contain negative entries

D. x and y are independent, since Σ is symmetric

Correct: B — $\sigma_x = 2$, $\sigma_y = 3$, tilted down-right

Why: The **diagonal** holds variances, so $\sigma_x = \sqrt{4}$, $\sigma_y = \sqrt{9}$ — A read them as standard deviations. $\rho = -3/(2 \cdot 3) = -0.5 < 0$: as x rises, y tends to fall — a down-right tilt. C: only the *diagonal* must be non-negative; off-diagonals may be negative freely (validity is positive semi-definiteness: here $\det = 36 - 9 = 27 > 0$, trace > 0 — L9's hand test says PD $\checkmark$). D: independence needs $\Sigma_{12} = 0$; symmetry is free.

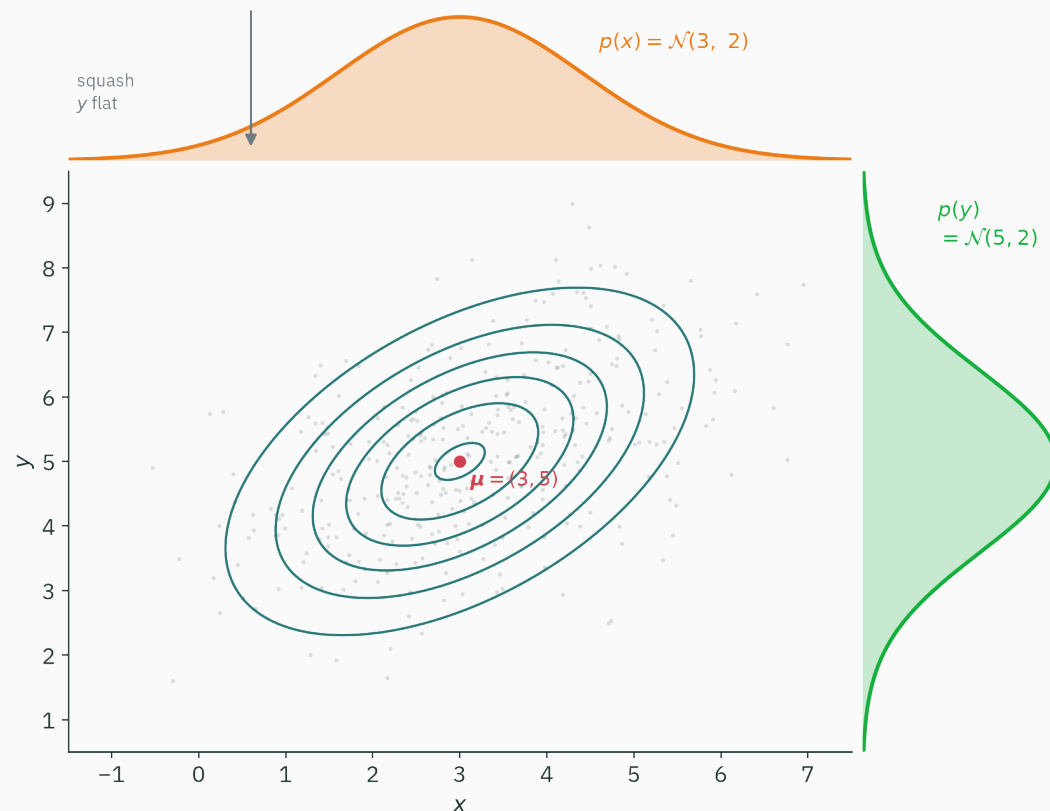
Slice and project: marginals and conditionals

Two questions you ask every joint

Your model is $p(x, y)$ over (height, weight). Two everyday questions:

1. “What’s the distribution of weight — ignoring height entirely?”
→ the **marginal** $p(y) = \int p(x, y) dx$: sum away what you don’t care about.
2. “What’s the distribution of weight **among 180 cm people**?”
→ the **conditional** $p(y \mid x = 180) = p(180, y)/p(180)$: freeze what you know, renormalize.

On the hill, these are **geometric moves**: marginalize = **project** (squash the hill flat), condition = **slice** (cut it with a plane). The next two figures are the whole story.



Squash the hill onto a wall — every y collapses into its x column.

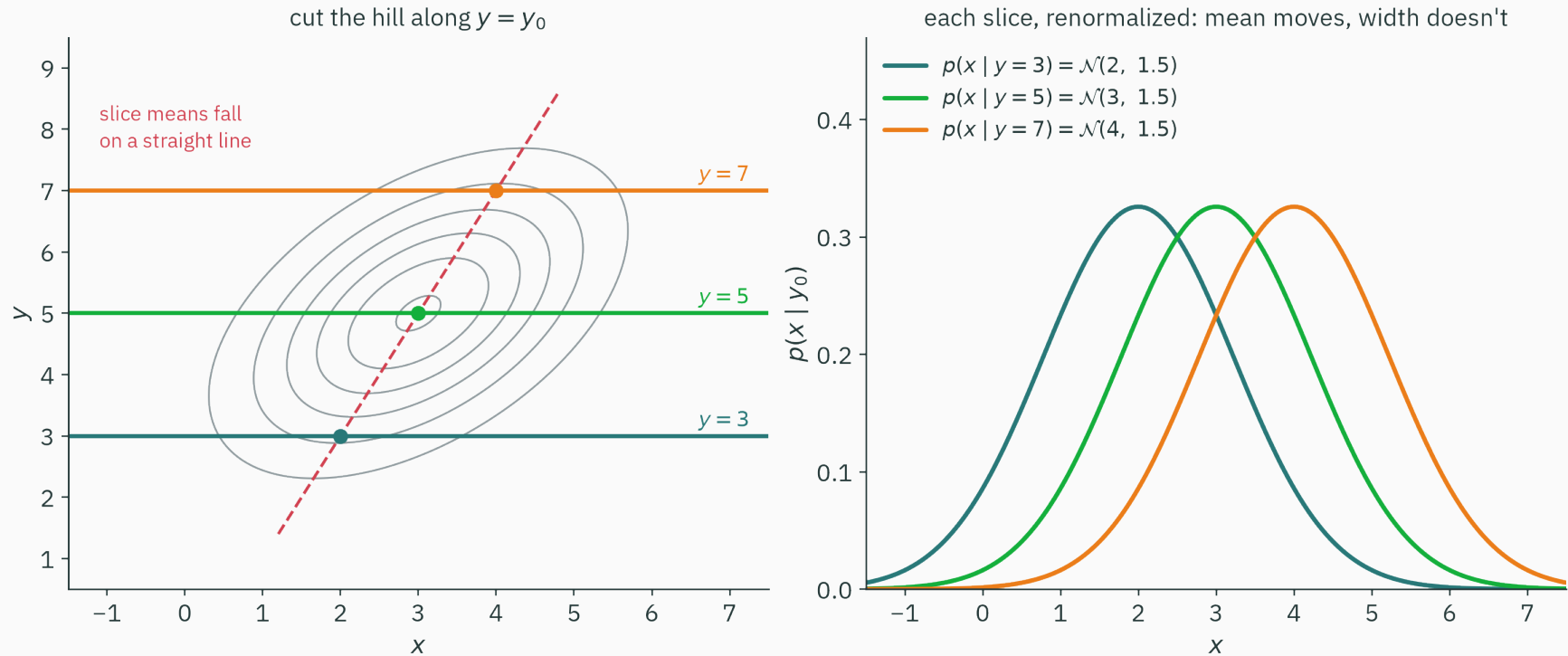
For our worked Gaussian:

$$p(x) = \mathcal{N}(3, 2)$$

$$p(y) = \mathcal{N}(5, 2)$$

Marginals: read them off μ and Σ — keep your entries, delete the rest. No integral needed.

Slice = condition



Cut along $y = y_0$: the profile revealed is $p(x, y_0)$ — a bump-shaped sliver.
Renormalize (divide by $p(y_0)$) and it becomes a legal 1-D density: $p(x | y = y_0)$.

The slice, quantified

Both panels of that figure obey two formulas (for 2-D; general d : same shapes, block form):

$$\mu_{x|y} = \mu_x + \frac{\Sigma_{12}}{\Sigma_{22}}(y_0 - \mu_y) \quad \sigma_{x|y}^2 = \Sigma_{11} - \frac{\Sigma_{12}^2}{\Sigma_{22}}$$

Our numbers ($\mu = (3, 5)$, $\Sigma_{12} = 1$, $\Sigma_{22} = 2$): observe $y_0 = 7 \rightarrow$

$$\mu_{x|7} = 3 + \frac{1}{2}(7 - 5) = 4 \quad \sigma_{x|y}^2 = 2 - \frac{1}{2} = 1.5 \quad (\text{matches the figure's orange slice})$$

- The mean **shifts linearly** with y_0 — the dashed line through the slice means is a preview of **regression** (L14's linear regression lives inside this picture).
- The variance **shrank** ($1.5 < 2$) and doesn't depend on y_0 at all — knowing y helps by the same amount everywhere. A distinctly Gaussian privilege.

Closure: every answer stayed in the family

operation on $\mathcal{N}(\mu, \Sigma)$	result
marginalize (project)	Gaussian — keep the relevant entries of μ, Σ
condition (slice)	Gaussian — mean shifts linearly, variance shrinks
linear map $Ax + b$	Gaussian — $\mathcal{N}(A\mu + b, A\Sigma A^\top)$
sum of independent Gaussians	Gaussian — means add, covariances add

The Gaussian family is closed under common linear operations used in ML, each with an explicit formula.

This closure is why **Kalman filters** track rockets in real time and **Gaussian processes** fit in a page of code: every update is “Gaussian in, Gaussian out”.
General block formulas: MML §6.5.

Code: check the slice by filtering samples

Don't trust the formula? Condition **empirically** — keep only samples where $y \approx 7$:

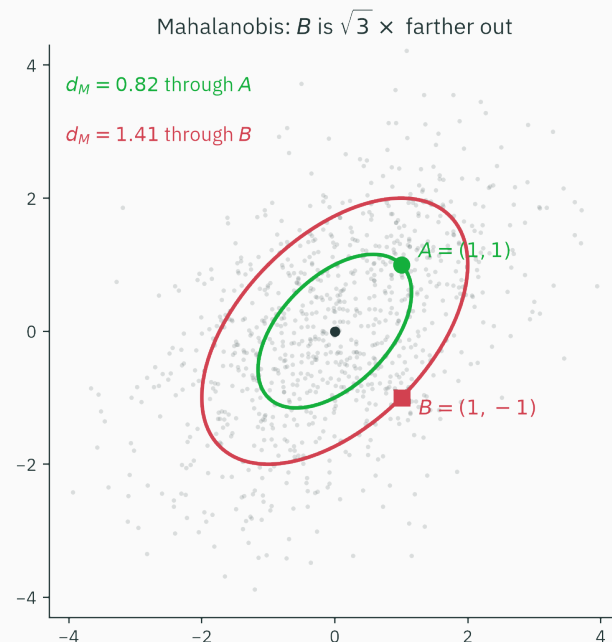
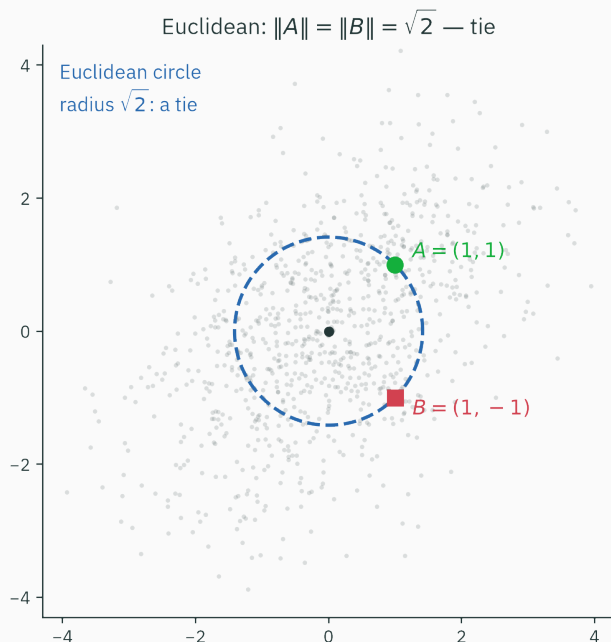
```
xs = d.sample((400_000,)).numpy()          # our N([3,5], [[2,1],[1,2]])
band = xs[np.abs(xs[:, 1] - 7.0) < 0.05]    # the thin slice y ≈ 7
band[:, 0].mean(), band[:, 0].var()
# (4.003, 1.497)      ← formula predicted N(4, 1.5) ✓
```

The filtered histogram **is** the renormalized slice. Simulation, geometry, and algebra tell one story.

Marginal = project. Conditional = slice. Both stay Gaussian — that's the spine of this lecture.

**Mahalanobis distance:
direction-aware σ 's**

Two points, same distance, different stories



$A = (1, 1)$ and $B = (1, -1)$ sit at **identical** Euclidean distance $\sqrt{2}$ from the mean — yet the cloud disagrees violently: A is in the thick of it, B is out in the cold.

Euclidean distance is **shape-blind**. We need a ruler that has seen Σ .

In 1-D, you already own the fix

L12's **z-score**: don't ask “how far?”, ask “**how many standard deviations?**”

$$z = \frac{x - \mu}{\sigma}$$

3 cm from the mean means nothing until you know σ

In d dimensions the question sharpens: standard deviations **in which direction?**
Along the cloud's long axis, spread is large — a big step is cheap. Across it, spread is small — the same step is expensive.

We need a distance that rescales every direction by that direction's own spread. Σ knows every direction's spread — so the ruler must be built from Σ .

Mahalanobis distance, defined

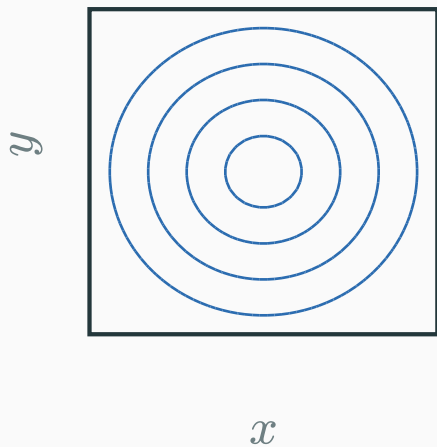
$$d_M^2(x) = (x - \mu)^\top \Sigma^{-1} (x - \mu)$$

- Why Σ^{-1} : dividing by variance, matrix-style — big-spread directions get **discounted**, tight directions get **amplified**.
- Sanity, $d = 1$: $d_M^2 = (x - \mu)^2 / \sigma^2$ — the z-score squared, exactly.
- Now look again at the Gaussian: $p(x) \propto \exp(-\frac{1}{2}d_M^2(x))$. **The exponent we dissected IS Mahalanobis distance.**

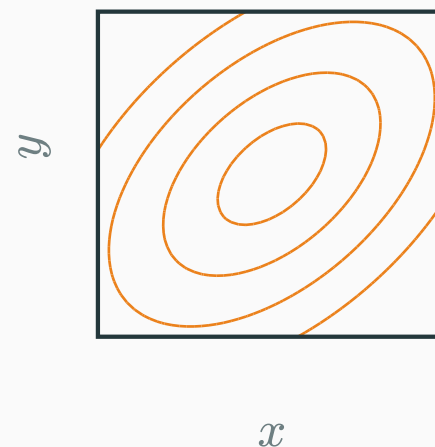
**Equal density $\Leftrightarrow$ equal Mahalanobis distance.
A Gaussian's contours are Mahalanobis “circles”.**

One ruler, two geometries

Euclidean $d^\top d$: circles



Mahalanobis $d^\top \Sigma^{-1} d$: ellipses



same four level values on both plots, $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ – the Mahalanobis unit ball is the covariance ellipse itself

Compute the conditional distribution

The slide inverts Σ live (chalkdust 1a.inv):

$$\Sigma^{-1} = \begin{pmatrix} 0.6667 & -0.3333 \\ -0.3333 & 0.6667 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

Score both suspects — hand-checkable in two lines each:

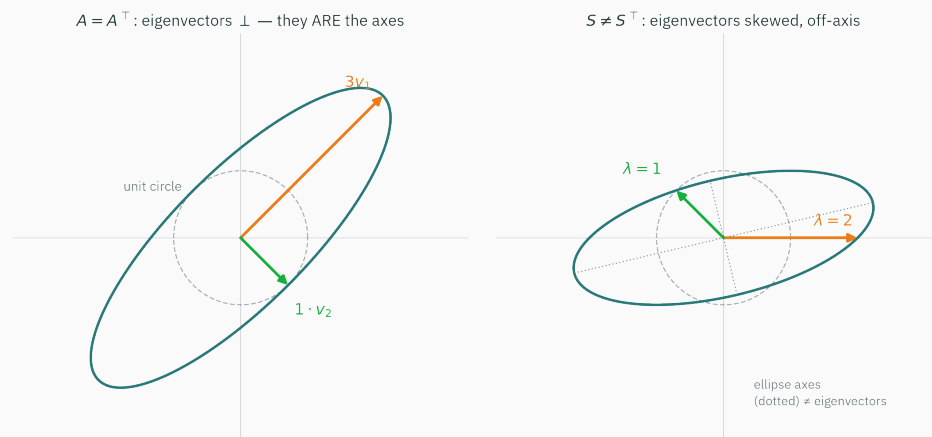
$$d_M^2(A) = (1, 1) \Sigma^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0.6667 \Rightarrow d_M(A) = 0.82 \text{ 'standard deviations'}$$

$$d_M^2(B) = (1, -1) \Sigma^{-1} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 2.0000 \Rightarrow d_M(B) = 1.41$$

Why exactly those numbers? A lies along $(1, 1)$ — the direction with variance 3. B lies along $(1, -1)$ — variance 1. Eigen-directions again... which brings us to the ★.

★ The ellipse is the
eigendecomposition of Σ

Claim. The contours of $\mathcal{N}(\mu, \Sigma)$ are ellipses centered at μ , with axes along the **eigenvectors of Σ** and semi-axes proportional to $\sqrt{\lambda_i}$.



L5's exact picture, IOU attached: “a Gaussian data cloud’s covariance eigenvectors will be the axes of the data ellipse.” Three slides of honest algebra pay it off.

★ Step 1: a contour is a level set of the exponent

D DERIVATION

Fix a density value c and take $\log$ of both sides of $p(x) = c$ — the normalizer is a constant, so it folds into the right-hand side:

$$p(x) = c \Leftrightarrow (x - \mu)^\top \Sigma^{-1} (x - \mu) = r^2 \quad (\text{some constant } r^2 \geq 0)$$

(exp is strictly monotone, so no information is lost taking the log — L2's log-space move, working geometry duty.)

Contours of the Gaussian = level sets of the quadratic form $d^\top A d$ with $A = \Sigma^{-1}$ — precisely L9's bowl. What remains: **which ellipses, pointed **where?****

★ Step 2: Σ and Σ^{-1} share eigenvectors

D DERIVATION

Let $\Sigma q = \lambda q$ (spectral theorem, L5: Σ symmetric $\rightarrow$ orthonormal q_i , real λ_i ; positive definite $\rightarrow$ all $\lambda_i > 0$, so Σ^{-1} exists). Multiply both sides by Σ^{-1} and divide by λ :

$$\Sigma q = \lambda q \quad \Rightarrow \quad q = \lambda \Sigma^{-1} q \quad \Rightarrow \quad \Sigma^{-1} q = \frac{1}{\lambda} q$$

Same eigenvectors, reciprocal eigenvalues. The bowl $d^\top \Sigma^{-1} d$ and the covariance Σ agree about the special directions — they only disagree about which one is “long”.

Where Σ has a **large** λ (much spread), Σ^{-1} has a **small** $1/\lambda$ (cheap distance) — the discount from the Mahalanobis slide, now with a mechanism.

★ Step 3: switch to the eigen-basis

D DERIVATION

L5/L9's signature move. Write $\Sigma = Q\Lambda Q^\top$, change coordinates $t = Q^\top(x - \mu)$ — a rotation onto the eigen-axes. The cross-terms die:

$$(x - \mu)^\top \Sigma^{-1} (x - \mu) = t^\top \Lambda^{-1} t = \frac{t_1^2}{\lambda_1} + \frac{t_2^2}{\lambda_2}$$

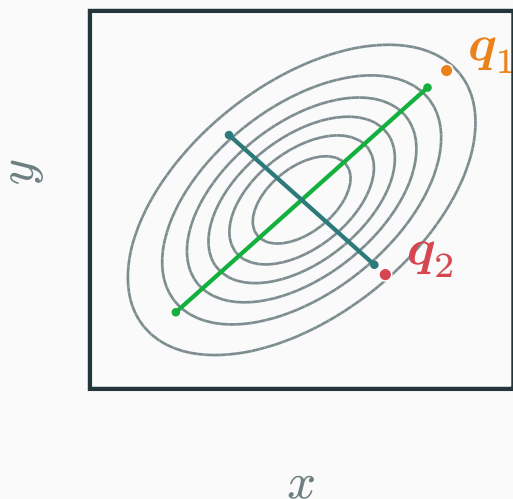
So the contour $= r^2$ reads, in eigen-coordinates:

$$\frac{t_1^2}{r^2 \lambda_1} + \frac{t_2^2}{r^2 \lambda_2} = 1 \quad \text{— the standard ellipse equation, semi-axes } r\sqrt{\lambda_1} \text{ and } r\sqrt{\lambda_2}$$

Contours of $\mathcal{N}(\mu, \Sigma)$: ellipses centered at μ , axes along the eigenvectors

q_i of Σ , semi-axes $\propto \sqrt{\lambda_i}$. ■

L5 planted it, L9 armed it, L13 proved it.



`la.eig-sym( $\Sigma$ )`, run by this slide:

$$\lambda_1 = 3, \quad \mathbf{q}_1 = \begin{pmatrix} 0.707 \\ 0.707 \end{pmatrix}$$

$$\lambda_2 = 1, \quad \mathbf{q}_2 = \begin{pmatrix} 0.707 \\ -0.707 \end{pmatrix}$$

Axis half-length ratio:

$$\sqrt{\lambda_1} : \sqrt{\lambda_2} = \sqrt{3} : 1 \approx 1.73 : 1$$

contours sampled from the density; axis segments drawn from `eig-sym`'s output, lengths $\propto \sqrt{\lambda_i}$ — computed, not sketched

Where this one fact keeps cashing out

- **PCA (L6), demystified**: PCA's principal directions are the eigenvectors of the data covariance — i.e., **the axes of this ellipse**. L6 found them by SVD; today you know what they mean probabilistically.
- **Whitening**: rotate by $Q^\top$, divide by $\sqrt{\lambda_i}$ — any Gaussian becomes $\mathcal{N}(0, I)$. Data preprocessing = ellipse $\rightarrow$ circle.
- **Optimization preview (L17)**: quadratic loss surfaces have these same elliptical contours; the ratio $\lambda_{\max}/\lambda_{\min}$ (how **stretched** the ellipse is) will decide how fast gradient descent can go.

One picture — eigenvectors of a symmetric matrix as ellipse axes — now runs statistics (today), compression (L6), and optimization (L17).

$\mathcal{N}(\mu, \Sigma)$ with $\Sigma = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$ has elliptical contours. Their **long** axis points along...

A. $(1, 0)$ — the first diagonal entry is largest

B. $(1, 1)$

C. $(1, -1)$

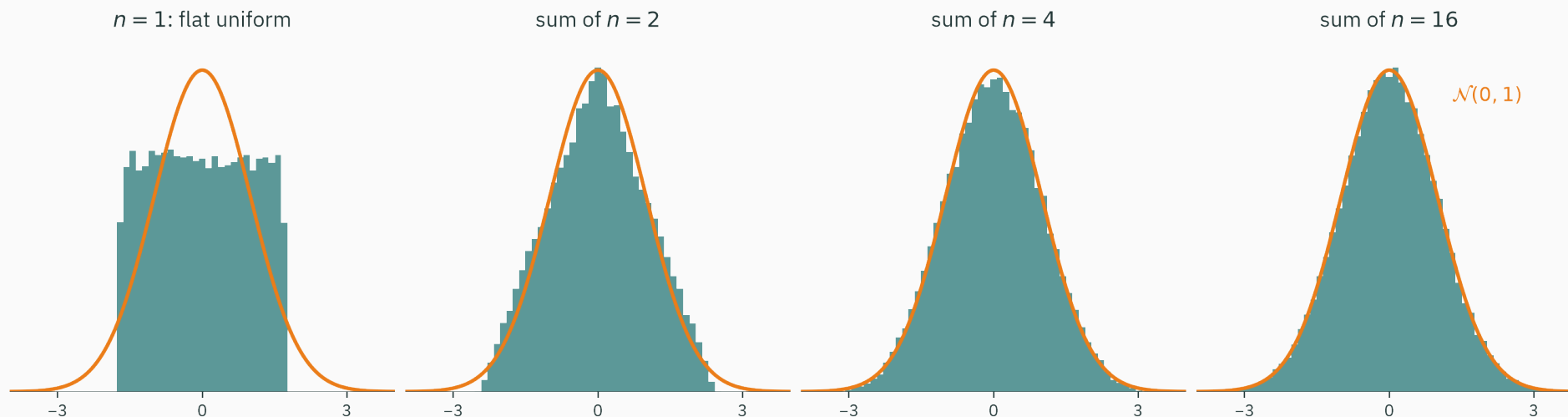
D. Can't say without knowing μ

Correct: B — along $(1, 1)$

Why: Same drill as our running matrix: for $\begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$, trace = 6 and det = 5 give $\lambda = 5, 1$ (L5's audit), with $(1, 1)$ the $\lambda = 5$ direction: most variance, longest axis ($\sqrt{5} : 1 \approx 2.2 : 1$). A: diagonal entries are per-**coordinate** variances — equal here (3 and 3), yet the ellipse is far from round; the tilt lives in the off-diagonal. D: μ only **slides** the ellipse; orientation is Σ 's alone.

Why the Gaussian is everywhere

Take the **least** Gaussian thing available — a flat uniform — and add independent copies:



By $n = 4$ the bell is unmistakable; by $n = 16$ the histogram and $\mathcal{N}(0, 1)$ are visually identical. Nothing about the uniform was special — dice, coin flips, exponentials all morph the same way.

The Central Limit Theorem, stated

Theorem (CLT, informal). If $X_1, \dots, X_n$ are i.i.d. with mean μ and finite variance σ^2 , then the standardized sum

$$\frac{X_1 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \rightarrow \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty$$

- The summands' own distribution **washes out** — only mean and variance survive the averaging.
- Sensor error, measurement jitter, and aggregate effects often combine many small independent contributions, so the central limit theorem motivates Gaussian noise models.

Honesty: we demoed the CLT and stated it; the proof needs tools (characteristic functions) beyond this course — see MML §6.4 for pointers. “Stated with pictures, cited, not proved” is our contract for it.

The Gaussian maximizes entropy at fixed variance

Fact (maximum entropy). Among **all** densities on $\mathbb{R}$ with a given mean μ and variance σ^2 , the Gaussian has the largest **entropy** — it is the least committed, most spread-out choice consistent with those two facts.

- Know only μ and Σ ? Then $\mathcal{N}(\mu, \Sigma)$ adds **zero** further assumptions. Any other choice quietly claims extra structure you never measured.
- “Entropy” — a number measuring non-commitment? Defining it properly is the whole business of **Module 5** (L22). This fact is planted today and **proved there**.

CLT: the Gaussian is where sums end up. Max entropy: it's where honesty starts. Two independent roads, one destination.

Summary of the Gaussian modelling assumptions

Why “Gaussian” in every ML paper:

property	modelling consequence	status
CLT	aggregated noise is born Gaussian	demoed today
max entropy	the least-assuming model given μ, Σ	stated $\rightarrow$ L22
closure	marginals, conditionals, sums: closed forms	seen today

GPT-2's init, diffusion noise, VAE latents, Kalman filters — each leans on at least two of the three.

**Born everywhere, presumes nothing, never punished by the algebra.
That's aristocracy.**

The take-home

Lecture 13 — summary

- **The MVN**: a bump on $\mathbb{R}^d$ with density $\propto \exp(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}))$ — the exponent is L9's quadratic form, $A = \Sigma^{-1}$.
- **Reading Σ** : diagonal = variances, off-diagonal = co-movement ($\rho = \Sigma_{12}/\sigma_1\sigma_2$); PSD because $\mathbf{a}^\top \Sigma \mathbf{a} = \text{Var}(\mathbf{a}^\top \mathbf{x}) \geq 0$.
- **Marginal = project, conditional = slice** — both stay Gaussian; conditional mean moves linearly, variance shrinks by a constant.
- **Mahalanobis** $d_M^2 = (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})$: “how many σ 's, direction-aware”; equal density $\iff$ equal d_M .
- ★ **The ellipse**: contour axes = eigenvectors of Σ , semi-axes $\propto \sqrt{\lambda_i}$ (L5, paid).
- **Why everywhere**: CLT + max entropy ($\rightarrow$ L22) + closure.

Read before L14 — MML §6.5–6.6 · CS229 probability notes handout. **Tutorial 6** this week: covariances by hand, slicing/projecting, the ellipse in NumPy.

Practice problems

Try on paper; check with NumPy (`np.cov`, `np.linalg.eigh`, `np.linalg.inv`).

1. Read $\Sigma = \begin{pmatrix} 9 & -5 \\ -5 & 4 \end{pmatrix}$: σ_x , σ_y , ρ , tilt direction. Then verify it **is** a valid covariance (L9 hand test: `det`, `trace`).
2. Why can no dataset ever have covariance $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$? (Hint: L9 classified this exact matrix. Find a direction a with “negative variance”.)
3. Compute the covariance of $(0, 0)$, $(2, 1)$, $(4, 2)$ by hand. What is $\det(\Sigma)$? What has the “ellipse” collapsed into, and which L4 concept is showing up?
4. For our running $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: compute d_M^2 of $(2, 2)$ and of $(-1, 1)$. Which is the outlier? Reconcile with the eigenvalues.
5. Sketch the 1-contour of $\mathcal{N}\left(\mathbf{0}, \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}\right)$: axis directions, axis-length ratio. (Trace/`det` shortcut welcome.)
6. For $\mu = (3, 5)$, $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: write $p(y \mid x = 1)$, then verify by sampling + filtering in code (as we did for $y = 7$).



Sampling: every Gaussian is a warped $\mathcal{N}(0, I)$

★ OPTIONAL

L12 taught the machine to make 1-D standard normals. Stack d of them: $z \sim \mathcal{N}(0, I)$ — a round cloud. How do we get $\mathcal{N}(\mu, \Sigma)$ from it?

Apply a linear map and use **closure**: $x = \mu + Az$ is Gaussian with mean μ and covariance

$$\text{Cov}(Az) = A \text{Cov}(z) A^\top = AA^\top$$

so we need a matrix with $AA^\top = \Sigma$ — a “square root” of Σ

The workhorse choice: the **Cholesky factor** L — lower-triangular, $LL^\top = \Sigma$, exists for every positive-definite Σ , cheap to compute.

$$z \sim \mathcal{N}(0, I), \quad x = \mu + Lz \quad \Rightarrow \quad x \sim \mathcal{N}(\mu, \Sigma)$$



Cholesky in code — and in pictures

★ OPTIONAL

```
Sigma = np.array([[2., 1.], [1., 2.]])
L = np.linalg.cholesky(Sigma)           # L @ L.T == Sigma
z = rng.standard_normal((100_000, 2))  # round cloud  N(0, I)
x = np.array([3., 5.]) + z @ L.T       # our Gaussian N(μ, Σ)
np.cov(x.T)                            # ≈ [[2.00, 1.00], [1.00, 2.00]] ✓
```

- Geometry: L maps the unit circle of z onto **the covariance ellipse** — L5's circle-to-ellipse picture was secretly a sampling algorithm.
- Another valid square root, straight from the ★: $\Sigma^{1/2} = Q\Lambda^{1/2}Q^\top$ — stretch by $\sqrt{\lambda_i}$ along the eigen-axes, rotate. Different A , same $AA^\top$, same distribution.
- This is precisely how `torch.distributions.MultivariateNormal` samples (it stores `scale_tril` — the Cholesky factor).

A Gaussian is a point and an ellipse.

Next: **Maximum Likelihood Estimation** — we can now describe data with distributions; next we *fit* them.